Center of Gravity Tracker for Fatigue Detection

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Summary:

Fatigue is a cause of serious accidents for excavator operators. A Center of Gravity (CoG) sensor is a promising new tool to non-invasively detect driver fatigue. In 2.75 Medical Device Design, we prototyped a research tool that tracks CoG from four sensors within the legs of a chair, and validated its accuracy and precision. Our contributions are, development of a low-cost integrated CoG detector for seated drivers and the design of a flexure structure to protect load cells from shocks, tension and shear forces. This device will allow researchers to develop operator's fatigue detection by tracking CoG.

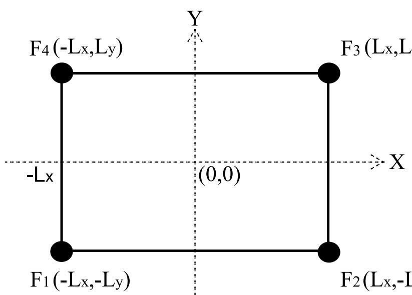
High Level Requirements:

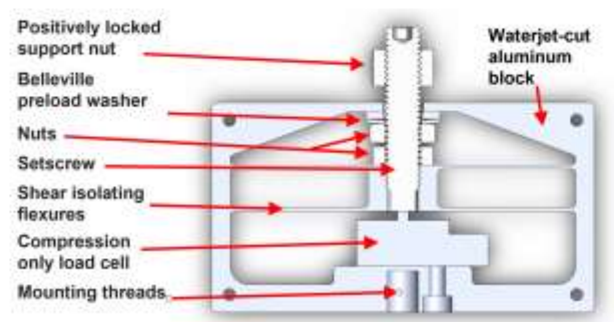
From discussions with SHI and in reviewing prior art the following requirements were identified to guide the design:

- Accuracy: The device can detect CoG of the operator with an absolute accuracy of better than +/- 2mm in X and Y.
- Robustness: The device can survive input forces from a 280lb driver in the worst case scenario loading conditions.
- Input: The device must take 12 V or 24 V DC input from a car battery.
- Output: The device should be able to record center of gravity at rate faster than 5 Hz. Raw reading from load cells and time should be recorded as well.
- Cost: The device should have a path towards commercialization and not involve exotic processes or materials.

Results:

As a team we designed, fabricated, and tested an integrated Center of Gravity detector shown below. Four load cells are protected by flexures and then installed into the base of a chair.





Our design successfully records the CoG of the operator, and stores the data locally on an SD card, streams it over USB, and displays it on a small LCD screen. The entire system is packaged within a robust plastic enclosure and uses industrial grade connectors. The resolution of the sensor is not limited by electrical noise, however in its current state it does not meet our accuracy requirement. Further calibration and analysis is required to address the lack of accuracy. The total BOM cost for a single unit is approximately $1,200 and can be further optimized and reduced at scale.

Impact:

We have created a purpose-built CoG detector that survives harsh loading without excessive cost.

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